

SF2940 Probability: Re-exam

Fall 2025

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Aids: One page hand written notes. No other aids are allowed. Formulary with distributions is included after the problem statement.

General instructions: Your solutions and answers should be legible, easy to follow and not lack in explanations. For full points a correct and well-presented reasoning is required.

Exam design: This exam consists of four independent problems. Each problem gives at most 5 points.

Problem 1. Let $X \sim \text{Po}(\lambda)$, and let Y be a random variables such that

$$Y | X = x \sim N(0, x) \quad \text{if } x > 0,$$

and $\mathbb{P}(Y = 0 | X = x) = 1$, if $x = 0$.

- (a) Find the characteristic function $\varphi_Y(t)$ of Y(2 points)
- (b) Show that $\frac{Y}{\sqrt{\lambda}}$ converges in distribution as $\lambda \rightarrow \infty$ and determine its limit.
.....(1 point)
- (c) For this part of the problem assume that λ is integer-valued. *Use the law of large numbers* to show that $\frac{X}{\lambda}$ converges in probability as $\lambda \rightarrow \infty$ and determine its limit..... (2 points)
(*Hint:* If $X_1 \sim \text{Po}(\lambda_1)$ and $X_2 \sim \text{Po}(\lambda_2)$ are independent, what is then the distribution of $X_1 + X_2$?)

Solution. (a) By the law of total expectation, we have that

$$\begin{aligned}\varphi_Y(t) &= \mathbb{E}[e^{itY}] = \mathbb{E}[\mathbb{E}[e^{itY} | X]] = \sum_{k=0}^{\infty} \mathbb{E}[e^{itY} | X = k] \mathbb{P}(X = k) \\ &= e^0 \mathbb{P}(X = 0) + \sum_{k=1}^{\infty} \varphi_{N(0,k)}(t) \mathbb{P}(X = k) \\ &= e^{-\lambda} + \sum_{k=1}^{\infty} e^{-\frac{1}{2}kt^2} \cdot e^{-\lambda} \frac{\lambda^k}{k!} \\ &= \sum_{k=0}^{\infty} e^{-\frac{1}{2}kt^2} \cdot e^{-\lambda} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^{-t^2/2})^k}{k!}\end{aligned}$$

$$= \exp(\lambda(e^{-t^2/2} - 1)).$$

(b) Since $\varphi_{aY}(t) = \varphi_Y(at)$ for all $a \in \mathbb{R}$, we have that

$$\varphi_{Y/\sqrt{\lambda}}(t) = \exp(\lambda(e^{-t^2/2\lambda} - 1)).$$

Recall that $e^{-t^2/2\lambda} = 1 - \frac{t^2}{2\lambda} + O\left(\frac{t^4}{4\lambda^2}\right)$, as $\lambda \rightarrow \infty$. Thus,

$$\begin{aligned} \varphi_{Y/\sqrt{\lambda}}(t) &= \exp\left(\lambda\left(1 - \frac{t^2}{2\lambda} - 1 + O\left(\frac{t^4}{4\lambda^2}\right)\right)\right) \\ &= \exp\left(-\frac{t^2}{2} + O\left(\frac{t^4}{4\lambda}\right)\right) \rightarrow e^{-t^2/2} = \varphi_{N(0,1)}(t) \quad \text{as } \lambda \rightarrow \infty. \end{aligned}$$

Thus, by the continuity theorem, we have that $\frac{Y}{\sqrt{\lambda}} \rightarrow N(0, 1)$ as $\lambda \rightarrow \infty$.

(c) Note that if $X_1 \sim \text{Po}(\lambda_1)$ and $X_2 \sim \text{Po}(\lambda_2)$ are independent, then

$$\varphi_{X_1+X_2}(t) = \varphi_{X_1}(t)\varphi_{X_2}(t) = e^{\lambda_1(e^{it}-1)}e^{\lambda_2(e^{it}-1)} = e^{(\lambda_1+\lambda_2)(e^{it}-1)} = \varphi_{\text{Po}(\lambda_1+\lambda_2)}(t).$$

In particular, $X_1 + X_2 \sim \text{Po}(\lambda_1 + \lambda_2)$. Consequently, we have that $X \stackrel{d}{=} Z_1 + \cdots + Z_\lambda$, where $Z_1, Z_2, \dots$ are independent $\text{Po}(1)$ -distributed random variables. Consequently, by the law of large numbers,

$$\frac{X}{\lambda} \stackrel{d}{=} \frac{\sum_{j=1}^{\lambda} Z_j}{\lambda} \xrightarrow{p} \mathbb{E}[Z_1] = 1.$$

Problem 2. Let $A \sim U(0, 1)$ and let X be a random variable such that

$$X \mid A = a \sim U(-a, a).$$

- (a) Determine $\mathbb{E}[A^2 \mid X]$(3 points)
 (b) Let $U = (X + A)/(2A)$. Are U and A independent? Thoroughly motivate your answer with complete computations..... (2 points)

Solution. (a) Note that since $X \mid A = a \in U(-a, a)$ and $A \sim U(0, 1)$, we have that

$$f_{X \mid A=a}(x) = \begin{cases} \frac{1}{2a} & \text{if } -a < x < a, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$f_A(a) = \begin{cases} 1 & \text{if } 0 < a < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Noting that by definition,

$$f_{X \mid A=a}(x) = \frac{f_{X,A}(x, a)}{f_A(a)},$$

we have that

$$f_{X,A}(x, a) = f_{X \mid A=a}(x) f_A(a) = \begin{cases} \frac{1}{2a} & \text{if } -1 < -a < x < a < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Consequently,

$$f_X(x) = \int_0^1 f_{X,A}(x, a) da = \int_{|x|}^1 \frac{1}{2a} da = \left[\frac{1}{2} \log a \right]_{|x|}^1 = -\frac{1}{2} \log |x| \quad \text{for } -1 < x < 1,$$

and 0 otherwise. Next, since

$$f_{A \mid X=x}(a) = \frac{f_{X,A}(x, a)}{f_X(x)} = \begin{cases} -\frac{1}{a \log |x|} & \text{if } |x| < a < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Thus, we have

$$\mathbb{E}[A^2 \mid X = x] = \int_{\mathbb{R}} a^2 f_{A \mid X=x}(a) da = \int_{|x|}^1 -\frac{a}{\log |x|} da = -\frac{1}{\log |x|} \left[\frac{a^2}{2} \right]_{|x|}^1 = \frac{|x|^2 - 1}{2 \log |x|}$$

and consequently,

$$\mathbb{E}[A^2 \mid X] = \frac{|X|^2 - 1}{2 \log |X|}.$$

(b) Let

$$U = \frac{X + A}{2A} \quad \text{and} \quad V = A.$$

Then, $A = V = h_2(U, V)$ and

$$X + A = 2AU \iff X = (2U - 1)A = (2U - 1)V = h_1(U, V).$$

Thus, we have that

$$|\mathbf{J}| = \begin{vmatrix} 2v & 2u-1 \\ 0 & 1 \end{vmatrix} = 2|v|.$$

Thus, by the transformation theorem,

$$f_{U,V}(u, v) = f_{X,A}((2u-1)v, v)2|v| = \begin{cases} \frac{1}{2v}2|v| & \text{if } -1 < -v < (2u-1)v < v < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Let us consider the support of $f_{U,V}$:

$$-v < (2u-1)v < v \iff -1 < 2u-1 < 1 \iff 0 < u < 1$$

Moreover, since $V = A$, we have that $0 < v < 1$, so that

$$f_{U,V}(u, v) = \begin{cases} 1 & \text{if } 0 < u < 1, 0 < v < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Consequently,

$$f_U(u) = \int_0^1 1 dv = 1 \quad \text{if } 0 < u < 1$$

and 0 otherwise. Hence, it follows that $f_{U,V}(u, v) = f_U(u)f_V(v)$ (since $f_V(v) = f_A(v)$), so that U and $V = A$ are independent.

Problem 3. Let $\mathbf{X} = (X_1, X_2, X_3, X_4, X_5)'$ be a multivariate normal random vector with mean vector $\boldsymbol{\mu}$ and covariance vector $\boldsymbol{\Lambda}$ given by

$$\boldsymbol{\mu} = \begin{pmatrix} 3 \\ 0 \\ 1 \\ 2 \\ -1 \end{pmatrix} \quad \boldsymbol{\Lambda} = \begin{pmatrix} 3 & 0 & -1 & 2 & 0 \\ 0 & 2 & 0 & 0 & 1 \\ -1 & 0 & 3 & 1 & 0 \\ 2 & 0 & 1 & 4 & 0 \\ 0 & 1 & 0 & 0 & 3 \end{pmatrix}.$$

- Let $Z_1 := X_1 - X_4 + X_5$ and $Z_2 := X_2 + 3X_3$. Determine $\mathbb{E}[Z_1 | Z_2]$ and the distribution of $Z_1 | Z_2 = 0$ (3 points)
- Construct two random vectors $\mathbf{Y}_1$ and $\mathbf{Y}_2$ such that each X_j , $j = 1, \dots, 5$, is a coordinate in either $\mathbf{Y}_1$ or $\mathbf{Y}_2$ and such that $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent. Determine the distributions of $\mathbf{Y}_1$ and $\mathbf{Y}_2$ (2 points)

Solution. (a) We have that

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = B\mathbf{X},$$

where

$$B = \begin{pmatrix} 1 & 0 & 0 & -1 & 1 \\ 0 & 1 & 3 & 0 & 0 \end{pmatrix}.$$

Thus,

$$\mathbb{E}\left[\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix}\right] = B\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 3 \end{pmatrix} =: \boldsymbol{\mu}_Z,$$

and

$$\text{Cov} \left(\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \right) = B\mathbf{\Lambda}B' = \begin{pmatrix} 6 & -5 \\ -5 & 29 \end{pmatrix} =: \mathbf{\Lambda}_Z.$$

Note that $\mathbb{E}[Z_1] = 0$, $\mathbb{E}[Z_2] = 3$, $\text{Var}(Z_1) = 6$, $\text{Var}(Z_2) = 29$ and $\text{Cov}(Z_1, Z_2) = -5$. As we have seen in the course, $Z_1 | Z_2 = z_2$ is normally distributed with expectation

$$\mathbb{E}[Z_1 | Z_2 = z_2] = \mathbb{E}[Z_1] + \frac{\text{Cov}(Z_1, Z_2)}{\text{Var}(Z_2)}(z_2 - \mathbb{E}[Z_2]) = -\frac{5}{29}(z_2 - 3)$$

and variance

$$\text{Var}(Z_1 | Z_2 = z_2) = \text{Var}(Z_1) \left(1 - \frac{\text{Cov}(Z_1, Z_2)^2}{\text{Var}(Z_1)\text{Var}(Z_2)} \right) = 6 \left(1 - \frac{25}{6 \cdot 29} \right) = 6 - \frac{25}{29} = \frac{149}{29}.$$

Thus, we have that

$$(Z_1 | Z_2 = 0) \sim N(15/29, 149/29)$$

and that

$$\mathbb{E}[Z_1 | Z_2] = -\frac{5}{29}(Z_2 - 3).$$

(b) Since $(X_i)_{i=1,\dots,5}$ are jointly normal, it follows that X_i and X_j , $i \neq j$, are independent if and only if $\text{Cov}(X_i, X_j) = 0$. By inspecting the covariance matrix, we see that X_1, X_3, X_4 are dependent on each other, but independent of X_2 and X_5 . Moreover, X_2 and X_5 are dependent. Thus, we let

$$\mathbf{Y}_1 = (X_1, X_3, X_4)' \quad \text{and} \quad \mathbf{Y}_2 = (X_2, X_5)'.$$

Then, we have that $\mathbf{Y}_1$ is a multivariate normal with mean vector $\boldsymbol{\mu}_1$ and covariance matrix $\mathbf{\Lambda}_1$ given by

$$\boldsymbol{\mu}_1 = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \quad \mathbf{\Lambda}_1 = \begin{pmatrix} 3 & -1 & 2 \\ -1 & 3 & 1 \\ 2 & 1 & 4 \end{pmatrix}.$$

and $\mathbf{Y}_2$ is a multivariate normal with mean vector $\boldsymbol{\mu}_2$ and covariance matrix $\mathbf{\Lambda}_2$ given by

$$\boldsymbol{\mu}_2 = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \quad \mathbf{\Lambda}_2 = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}.$$

Problem 4. Let $X_1, X_2, \dots$ be independent $U(0, 1)$ -distributed random variables. For each $t > 0$, define the random variable

$$N(t) := \min\{n \geq 1 : X_n \leq t\},$$

that is, for any given $t > 0$, $N(t)$ is the smallest n such that $X_n \leq t$. (For example, $\{N(t) = 3\} = \{X_1 > t\} \cap \{X_2 > t\} \cap \{X_3 \leq t\}$.)

- Determine the probability mass function $p_{N(t)}(k)$ of $N(t)$ (2 points)
- Let $S_n = \sum_{j=1}^n X_j$. Show that $S_{N(t)} \xrightarrow{P} X_1$ as $t \rightarrow 1$ (1 point)
- Determine $\mathbb{E}[S_{N(t)}]$ (2 points)
(Hint: Use the law of total expectation and the formulas below.)

The following formulas might be helpful:

For a random variable Y and an event A , we have that

$$\mathbb{E}[Y | A] = \frac{\mathbb{E}[Y \mathbf{1}_A]}{\mathbb{P}(A)}.$$

For $r \in (0, 1)$, we have that

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \quad \text{and} \quad \sum_{n=1}^{\infty} n r^{n-1} = \frac{1}{(1-r)^2}.$$

Solution. (a) Note that $\{N(t) = n\} = \{X_1 > t\} \cap \cdots \cap \{X_{n-1} > t\} \cap \{X_n \leq t\}$. Thus, by independence of $X_1, X_2, \dots$, we have that

$$\begin{aligned} p_{N(t)}(k) &= \mathbb{P}(N(t) = k) = \mathbb{P}(\{X_1 > t\} \cap \cdots \cap \{X_{n-1} > t\} \cap \{X_n \leq t\}) \\ &= \mathbb{P}(X_1 > t) \cdots \mathbb{P}(X_{n-1} > t) \mathbb{P}(X_n \leq t) = t(1-t)^{n-1}. \end{aligned}$$

(b) Fix $\epsilon > 0$. We have that

$$\mathbb{P}(|S_{N(t)} - X_1| > \epsilon) \leq \mathbb{P}(S_{N(t)} \neq X_1) = \mathbb{P}(N(t) \geq 2) = \mathbb{P}(X_1 \geq t) = t \rightarrow 1 \quad \text{as } t \rightarrow 1.$$

Since $\epsilon > 0$ was arbitrary, we have that $S_{N(t)} \xrightarrow{p} X_1$.

(c) We have that

$$\begin{aligned} \mathbb{E}[S_{N(t)}] &= \mathbb{E}[\mathbb{E}[S_{N(t)} | N(t)]] = \sum_{n=1}^{\infty} \mathbb{E}[S_{N(t)} | N(t) = n] \mathbb{P}(N(t) = n) \\ &= \sum_{n=1}^{\infty} \mathbb{E}[S_n | N(t) = n] \mathbb{P}(N(t) = n) \end{aligned}$$

Note that $X_1, X_2, \dots$ and $N(t)$ are dependent. We have that

$$\begin{aligned} \mathbb{E}[S_n | N(t) = n] &= \mathbb{E}\left[\sum_{j=1}^n X_j \mid N(t) = n\right] = \sum_{j=1}^n \mathbb{E}[X_j | N(t) = n] \\ &= \sum_{j=1}^{n-1} \mathbb{E}[X_j | X_j > t] + \mathbb{E}[X_n | X_n \leq t]. \end{aligned}$$

Next, we note that (recall the hint)

$$\mathbb{E}[X_j | X_j > t] = \frac{\mathbb{E}[X_j \mathbf{1}_{\{X_j > t\}}]}{\mathbb{P}(X_j > t)}$$

where

$$\mathbb{E}[X_j \mathbf{1}_{\{X_j > t\}}] = \int_t^1 x f_{X_j}(x) dx = \int_t^1 x dx = \left[\frac{x^2}{2}\right]_t^1 = \frac{1-t^2}{2}$$

and $\mathbb{P}(X_j > t) = 1 - t$, so that

$$\mathbb{E}[X_j | X_j > t] = \frac{1-t^2}{2(1-t)} = \frac{1+t}{2}$$

and similarly, since

$$\mathbb{E}[X_n \mathbf{1}_{\{X_n \leq t\}}] = \int_0^t x f_{X_n}(x) dx = \int_0^t x dx = \left[\frac{x^2}{2}\right]_0^t = \frac{t^2}{2}$$

and $\mathbb{P}(X_n \leq t) = t$, we have that

$$\mathbb{E}[X_n | X_n \leq t] = \frac{t}{2}.$$

Consequently,

$$\mathbb{E}[S_n | N(t) = n] = (n-1) \frac{1+t}{2} + \frac{t}{2}$$

and thus

$$\begin{aligned} \mathbb{E}[S_{N(t)}] &= \sum_{n=1}^{\infty} \left((n-1) \frac{1+t}{2} + \frac{t}{2} \right) t(1-t)^n \\ &= \frac{t(1+t)}{2} \sum_{n=1}^{\infty} (n-1)(1-t)^n + \frac{t^2}{2} \sum_{n=1}^{\infty} (1-t)^n \\ &= \frac{t(1+t)(1-t)^2}{2} \sum_{n=1}^{\infty} (n-1)(1-t)^{n-2} + \frac{t^2}{2} \sum_{n=1}^{\infty} (1-t)^n \\ &= \frac{t(1+t)(1-t)^2}{2} \sum_{n=1}^{\infty} n(1-t)^{n-1} + \frac{t^2}{2} \sum_{n=1}^{\infty} (1-t)^n \end{aligned}$$

By the provided formulas, we have that

$$\sum_{n=1}^{\infty} (1-t)^n = \sum_{n=0}^{\infty} (1-t)^n - 1 = \frac{1}{1-(1-t)} - 1 = \frac{1}{t} - 1,$$

and that

$$\sum_{n=1}^{\infty} n(1-t)^{n-1} = \frac{1}{(1-(1-t))^2} = \frac{1}{t^2}.$$

So that

$$\begin{aligned} \mathbb{E}[S_{N(t)}] &= \frac{t(1+t)(1-t)^2}{2} \frac{1}{t^2} + \frac{t^2}{2} \left(\frac{1}{t} - 1 \right) = \frac{(1+t)(1-t)^2}{2t} + \frac{t}{2} - \frac{t^2}{2} \\ &= \frac{1-t-t^2+t^3}{2t} + \frac{t}{2} - \frac{t^2}{2} = \frac{1-t}{2t} = \frac{1}{2} \left(\frac{1}{t} - 1 \right). \end{aligned}$$

Good luck!

Distribution	Notation	Probability function	$\varphi_X(t)$
One point	$\delta(a), a \in \mathbb{R}$	$p(a) = 1$	e^{ita}
Bernoulli	$\text{Be}(p); 0 \leq p \leq 1$	$p(0) = q, p(1) = p; q = 1 - p$	$q + pe^{it}$
Binomial	$\text{Bin}(n, p), n = 1, 2, \dots; 0 \leq p \leq 1$	$p(k) = \binom{n}{k} p^k q^{n-k}, k = 0, 1, 2, \dots, n; q = 1 - p$	$(q + pe^{it})^n$
Geometric	$\text{Ge}(p), 0 \leq p \leq 1$	$p(k) = pq^k, k = 0, 1, 2, \dots; q = 1 - p$	$\frac{p}{1 - qe^{it}}$
Poisson	$\text{Po}(\lambda), \lambda > 0$	$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, 2, \dots$	$e^{\lambda(e^{it} - 1)}$

TABLE 1. Discrete distributions

Distribution	Notation	Density	$\varphi_X(t)$
Uniform	$U(a, b), a < b$	$f(x) = \frac{1}{b-a}, a < x < b$	$\frac{e^{itb} - e^{ita}}{it(b-a)}$
Exponential	$\text{Exp}(\lambda), \lambda > 0$	$f(x) = \frac{1}{\lambda} e^{-x/\lambda}, x > 0$	$\frac{1}{1 - \lambda it}$
Laplace	$L(\lambda), \lambda > 0$	$f(x) = \frac{1}{2\lambda} e^{- x /\lambda}, x \in \mathbb{R}$	$\frac{1}{1 + \lambda^2 t^2}$
Beta	$\beta(r, s), r, s > 0$	$f(x) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} x^{r-1} (1-x)^{s-1}, 0 < x < 1$	*
Normal	$N(\mu, \sigma^2), \mu \in \mathbb{R}, \sigma > 0$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2}, x \in \mathbb{R}$	$e^{i\mu t - \frac{1}{2}\sigma^2 t^2}$
Standard normal	$N(0, 1)$	$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}, x \in \mathbb{R}$	$e^{-\frac{1}{2}t^2}$
Log-normal	$\text{LN}(\mu, \sigma^2), \mu \in \mathbb{R}, \sigma > 0$	$f(x) = \frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\log x - \mu)^2/\sigma^2}, x > 0$	*
Cauchy	$C(m, \eta), \eta > 0, m \in \mathbb{R}$	$f(x) = \frac{1}{\pi} \frac{\eta}{(x-m)^2 + \eta^2}, x \in \mathbb{R}$	$e^{imt - \eta t }$

TABLE 2. Continuous distributions